

Case Study

The implementation of lean and digital management techniques using artificial intelligence in industrial settings

Aleksey Grigorievich Tashkinov¹

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Abstract

This article proposes an interdisciplinary approach to analyzing the problems associated with the introduction of digital technologies into industry, with special attention to artificial intelligence. The author offers a methodological framework based on the principles of lean manufacturing and artificial intelligence to eliminate the identified gaps. To improve the efficiency of production systems, a model is proposed that combines the principles of lean manufacturing with digital tools such as artificial intelligence. The introduction of a new concept and method of production process management requires a revision of the project management strategy and organizational practices. As one of the main results, a project is proposed aimed at introducing lean and digital production methods using modern digital technologies. The introduction of digital technologies has led to the appearance of preliminary results that allow the heads of production departments to make informed decisions based on up-to-date data. Enterprises using the potential of artificial intelligence demonstrate higher efficiency, complete orders ahead of schedule and benefit customers through innovative software and algorithms. This approach allows organizations to effectively manage their activities, providing value to customers and increasing their satisfaction. Issues related to Industry 4.0, digitalization and artificial intelligence play a key role in the context of business transformation, therefore, the study of these aspects can be valuable both for the academic community and for the practical activities of companies.

Keywords Artificial intelligence · Digital technology · Industry 4.0 · Lean manufacturing · Project management

Abbreviations

AI	Artificial intelligence
DL	Deep learning
ML	Machine learning
5S	System 5
I.4.0.	Industry 4.0.
LM	Lean management
MIT	Massachusetts Institute of Technology's
LEAN	Lean manufacturing, production
IoT	Internet of Things
DT	Digital technologies
SCADA	Supervisory control and data acquisition
LPSS	Lean-based production systems

✉ Aleksey Grigorievich Tashkinov, alekss.perm@gmail.com | ¹Candidate of Economic Sciences, Department of Economics and Industrial Production Management, Perm National Research Polytechnic University, 29 Komsomolsky Prospekt, Perm Krai, Perm 614990, Russia.



HTO	Human-technology-organization model
JIT	Just in time
SMED	Single minute exchange die
VSM	Value stream mapping

1 Introduction

The twenty-first century has seen a transformative change in the industrial landscape, driven largely by the rapid advancement of artificial intelligence (AI). This shift is often referred to as the "Fourth Industrial Revolution" or "Industry 4.0", which focuses on digital connectivity, automation, and intelligent decision-making [1]. The business landscape is experiencing intense changes due to deep transformations caused by temporary and structural factors, such as global market integration, technological advancements, and especially digitalization. These changes are more than just the introduction of new tools based on information and communication technologies (ICTs) into organizational structures and processes.

Quality is achieved thanks to people, their attitudes, and their commitment [2]. The tools and technical skills are necessary. However, studies have shown that soft skills are more important for project's success [3]. The starting point for digitalization development work is to understand the overall current digitalization status of the organization and thus, create an overall picture of its strengths and weaknesses and possible improvement actions for the future [4]. Thus, digitalization can mean enhancing existing processes, finding new opportunities within existing business domains, or finding new opportunities outside existing business domains [5]. The term project management approach is the "highest level of abstraction used when describing how a project will be designed" [6], whereas a project management methodology is more granular and provides specific guidance about how to manage a project [7–9].

The ubiquity and rapid advancement of digital technologies have resulted in a constant flux in consumer preferences and market dynamics. These transformations, in turn, have imposed new expectations on organizations. Consequently, an increasing number of companies are leveraging digital tools to optimize customer experiences and enhance their value propositions through data-driven insights. Nevertheless, enterprises are also confronted with economic and geopolitical uncertainties arising from the ever-evolving technological landscape. This may lead many organizations to naturally gravitate towards a more conservative approach.

The dilemma arises: should businesses integrate digital logic into their core operations, or should they continue to view it as an adjunct to their business processes?

The process of digital transformation is a critical aspect for all organizations, irrespective of their sector. Digitalization has become an indispensable requirement and a key driver of competitiveness. Lean management, a proven methodology for optimizing production systems, can also be effectively applied to enhance efficiency. As Industry 4.0 advances and becomes more prevalent, there is a growing emphasis on digitalization initiatives aimed at improving business outcomes and boosting productivity. Organizations are actively pursuing initiatives to optimize their processes and increase efficiency. However, some implementations of Industry 4.0 technologies within enterprises may lack proper coordination, raising questions about their alignment with established methods like lean management, which have demonstrated their effectiveness in achieving organizational goals. Therefore, this study seeks to delve into the synergy and practical application of Industry 4.0 and lean manufacturing in order to enhance enterprise efficiency and prevent conflicting actions during the implementation of these two initiatives. The primary objective of this investigation is to conduct a comprehensive analysis of the current state of efficiency within companies and propose strategies that can significantly boost productivity through an integrated approach, combining the principles of Industry 4.0, lean manufacturing, digital production, and artificial intelligence. These recommendations are founded on the integration of lean management models with the concepts of Industry 4.0, as well as the application of theoretical principles from both management methodologies in practical settings. Through this research, we aspire to contribute to the enhancement of production efficiency and overall company performance.

Based on the aforementioned exposition of the issue, we propose a framework for the paper that will predominantly elucidate the crux of our investigation.

The manuscript is divided into several main sections. The *first* part is an introduction, which emphasizes the importance of scientific research and outlines the topic under study. This section also provides a brief overview of the theoretical and methodological foundations of the research and identifies gaps in the current state of knowledge on the topic. The *second* part of the manuscript focuses on terminology related to various concepts such as lean manufacturing, digital manufacturing, Industry 4.0, artificial intelligence, machine learning, and other relevant terms. This section provides a

critical analysis of literature on the use of AI and identifies challenges in this area. It also offers recommendations for future research in order to address these challenges and contribute to the advancement of knowledge in the field. The *third* section explores the complexities of conducting research to optimize the implementation of digital technologies within the context of Industry 4.0. It examines the main methodologies employed in the study of specific cases, as well as the process of enhancing the production system based on lean manufacturing principles and digital technologies. The *fourth* section presents a comprehensive analysis of the findings, including the outcomes achieved during the process and a strategy to improve production efficiency through the utilization of artificial intelligence in business management. This part examines the implementation of the suggested approach utilizing digital technologies, with examples from the realm of digital manufacturing, with special attention given to their integration within the framework of Industry 4.0. Furthermore, a practical demonstration of how the model can be applied to meet the specific requirements of an organization is provided. We describe the process of implementing the method and offer recommendations for future research and development. In conclusion, we summarize the findings of our study and draw key conclusions. Overall, we aim to outline the direction of scientific and practical inquiry in this field.

2 Literature review

2.1 The concept of lean management

The concept of lean management (LM) was a response to the emergence of strong competition with severe resource restrictions [10]. LM is a generic term that was coined by the Massachusetts Institute of Technology's (MIT) research program titled the "International Motor Vehicle Programme", which was launched in 1979 and delivered groundbreaking results 11 years later, when James P. Womack et al. published their findings in "The Machine that Changed the World" [11]. The founder of the concept of LEAN (lean manufacturing) is Taiichi Ono, who created the production system of Toyota Corporation in the 1950s [12]. Their research focused on the performance gap between the Japanese automotive industry and the Western industry. The basis of their data was an analysis of 52 assembly plants in 14 countries. The five-year, five-million-dollar program revealed that Japanese plants were significantly more productive, because of their production and management style, which the authors called "Lean". Most notably, even still, the car manufacturer Toyota was at the center of attention, as they have, arguably, contributed the most to what we know today as lean. Initially, lean referred only to the manufacturing part, but later, the term lean also included the management style. Today, there are multitude of lean derivatives, such as lean enterprises, lean leadership, lean logistics, lean production, lean organization, lean thinking, lean service, and lean planning [13].

Thus, the works of Dennis P. Hobbs, J. Womack, Liker J.K. Morgan J, Taiichi Ono, Rother M., and Tanaka T consider various aspects of the introduction of the concept of LEAN in enterprises [10–12]. Womack, Jones, and Roos coined the term "lean production" in their 1990 book "The Machine that Changed the World" [11]. To describe the manufacturing paradigm established by the Toyota Production System.

There is also criticism around "The Machine that Changed the World". Womack et al. claimed that the cultural and societal context was not the decisive factor for the superior performance of Japanese manufacturers: thus, we devote our attention in the future to a careful explanation of the logic and techniques of lean production [13]. This is thought-provoking, as the researchers already knew about Taiichi Ohno's book, who is said to be the coinventor the Toyota Production System (which stands for LEAN more than any other system), where he stresses that everything depends on the will and wanting of the employees [13, pp. 91–92]. On the other hand, it is understandable that it is much easier to research and describe hard facts than it is to grasp complex human aspects. Mike Rother boiled it down to the essence. This is because an organization's collection of practices and principles at any point in time is an outcome that springs from its members' routines of thinking and behavior. An organization's competitiveness, ability to adapt, and culture arise from the routines and habits by which the people in the organization conduct themselves every day. It is an issue of human behavior [14, p. 16].

LEAN is therefore called lean, which allows you to do more and more, and spend less and less human effort, less equipment, less time and less production space—while at the same time approaching providing the consumer with exactly what he wanted [15, p. 33].

It is believed that LEAN ideas came to Russia in 2004, together with foreign companies that opened factories and branches in the country, or organized joint production with Russian enterprises on the basis of industrial cooperation. Each company brought to our practice those approaches to the implementation of LEAN, which are accepted in her

country (Toyota, Nissan, Ford, etc.). Studying the experience of foreign colleagues, Russian enterprises began to create their production systems on the basis of common approaches to LEAN, introducing positive aspects from the systems of rationalization and invention, scientific organization of labor, etc.

According to J. Womack, "Lean production is a breakthrough approach to management and quality management, ensuring long-term competitiveness without a substantial investment based on the principle of universal reduction of possible losses, involving each employee in the process of business optimization" [13, p. 32].

According to D. Hobbs, "Lean production is a technique aimed at constantly reducing operating costs. Its task is to develop and implement a production line capable of producing different types of products at the exact time that is necessary for this purpose [16, p. 33].

Each view reflects an important aspect of the concept under analysis and has every right to exist. In this case, in our opinion, this definition requires further reflection and addition. Taking into account the existing views of scientists, the author proposes to clarify what is meant by the concept of LEAN as a supplement. The concept of LEAN refers to a strategy based on the principles of a systematic approach to determining simple solutions for identifying and eliminating hidden losses (production of goods/services), which allows optimizing processes to improve the decisive performance of the enterprise: such as cost, quality, service and time; giving the consumer new competitive advantages; creating added value for customers; and meeting their needs. The idea is to constantly strive to eliminate any type of loss, with the involvement of each employee in the procedure [17].

In general, the concept of LEAN can be considered a special mechanism for modernizing technological processes at the enterprise, which makes it possible to reduce costs at the enterprise level due to the introduction of these principles without attracting additional resources related to the purchase of advanced technologies, modern equipment, etc. [18–20].

The strengths of this LEAN concept are due to three factors:

1. This is a proven structure that provides forms and techniques that allow you to quickly obtain results (the method assumes the presence of losses and "attacks" losses without studying them, which is almost always extremely effective).
2. The approach works well at the level of ordinary employees, where there are a small number of small and minor losses (invisible to most managers).
3. Quickly gives actual results.

In order to fully understand the development path of the Toyota production system, it is necessary to analyze the key stages of its evolution. Like many other companies, Toyota has come to realize that excessive attention to tools is not the optimal approach [10, 21]. Toyota currently applies lean manufacturing principles that are significantly different from the traditional tool-based approach. In lean manufacturing, strategic principles and a common vision are central, not individual tools. This approach is based on Hoshina Kanri's strategy, which became the starting point for the development of Toyota's production system. Toyota's production system, starting with a set of tools, has evolved into a complex and efficient system [22]. This article examines the concept of lean manufacturing, which involves a continuous commitment to adapt to the needs of customers, employees and external conditions. This approach is called kaizen and is a continuous process of improvement. Employees make suggestions for improving their work, which are then implemented. The most important components of the lean manufacturing methodology are collaboration and teamwork. The concept of gemba, or "production site", emphasizes the need for physical presence in the workplace, rather than remote control.

The fundamental principles of lean manufacturing, such as the constant pursuit of excellence and respect for human dignity, were developed in the Toyota production system in the 1930s and 1950s. The founders of this system, Sakichi Toyoda and his son Katsuhiko, made significant contributions to the formation of these ideas. These principles are rooted in Samuel Smiles' book *Self-Help*, which served as a source of inspiration for the first lean manufacturing specialists. The principles of lean manufacturing are at the heart of the Toyota Way, a complex system of principles that is difficult to describe briefly, as Jeff Liker is trying to do [11]. But ultimately, it is thinking people who lay the foundation for sustainable success in any business. The concept of lean manufacturing is not just a methodology for optimizing production processes, but also a comprehensive approach to business management. It covers all aspects of the company's business, including accounting, information technology, human resources management, marketing, sales, procurement, distribution, design and development.

Creating value and eliminating waste are equally important within the framework of the concept of lean manufacturing. Some experts believe that the Toyota production system (TPS) should not be considered as a "Toyota

production system", but as a "system of thinking people", meaning that it is based on the principles of thinking and decision-making aimed at continuous improvement and optimization of all aspects of the company's activities.

The implementation of the Lean concept can be used in two ways. In the first case, a set of one-time organizational and technical measures assumes that it is used in the reengineering of business processes in accordance with the methodology of M. Hammer [23]. In the second case, the creation of lean production actually means the development of known elements of the constant improvement of the kaizen system, which belongs to M. Imai, an ideologist of one of the most effective models for implementing the concept of lean production [24].

Kaizen in business is a constant improvement, from the workplace to senior management [15]. The main goal of kaizen is lossless production. Companies using the LEAN concept apply the principle of creating cross-functional commands (command formation). For success Considerable attention, interest, planning and communication are required from the top managers of the team in the enterprise. Every organizational project, according to D. Tepping, T. Shuker, should begin with the "Cath-ball" procedure—throwing the ball [25, pp. 26–27].

We propose considering the basic principles of LEAN [26]:

1. Principle—Bottom-Up Control
2. Principle—Inventory reduction
3. Principle—Continuity of transformations

The LEAN implementation project is not complete, and the period of stability of any process is as follows. We summarize the main provisions of the LEAN concept and its role in ensuring the efficiency of engineering enterprises, as noted in our study:

The LEAN concept uses a project approach to solve specific tasks by means of initiative working groups and specialized departments.

The LEAN concept is focused on a process approach to managing enterprise improvement.

1. To expand the work on the continuous improvement of production processes, control bodies are created that implement and monitor the decisions made.
2. To ensure the unity of approaches to the implementation of LEAN tools in practice, the company developed a set of provisions and regulatory documents regulating the activities of various departments.

Thus, the effectiveness of the LEAN concept in plants and factories is partly because many of the tools of this method greatly contribute to ensuring concentration, structure and discipline. When properly applied, it also develops a strong sense of ownership among ordinary workers.

On this basis, LEAN is a single set of extremely efficiently selected tools that can be applied to the production process. In addition, although the basic philosophy of LEAN (elimination of losses, cells, pulling instead of ejecting) refers to any work, its standard tools, applied without significant modification not to production but to office procedures, are much less effective. With this in mind, to obtain a synergistic effect and maximum results, the concept of LEAN should be introduced using the concepts of fast-reacting and active production.

The concept of lean manufacturing embodies a holistic approach that is underpinned by an integrated framework of principles and practices aimed at enhancing the operational effectiveness of an organization, encompassing its production lines and machinery.

The core objective of this methodology is to eradicate all forms of waste by streamlining processes and eliminating those that fail to contribute to value generation, with the active involvement of all personnel.

The implementation of lean manufacturing practices enables a substantial enhancement in production efficiency, resulting not only in increased output, improved product quality, and minimized waste, but also necessitates the involvement of advanced artificial intelligence expertise.

Next, consider the following concept that meets the modern requirements for ensuring the competitiveness of enterprises: the concept of Industry 4.0.

Implementation concepts should be well aligned with an integrated approach to production system management when implementing Industry 4.0 lean production management.

2.2 Development concepts "Industry 4.0"

The business landscape is undergoing periods of rapid change due to significant digital transformations, caused by various factors such as global market integration and technological advancements, particularly digitalization. This change is not only related to the introduction of traditional lean and digital manufacturing tools into organizations, but also to the adoption of new digital tools powered by artificial intelligence.

German consulting firms at the CeBIT trade show in Hanover in 2013 coined the term "Economy 4.0" or "Industry 4.0" to describe this new economic landscape. This term refers to the combination of economic, business and social factors that arise from the emergence and development of various technologies interconnected through the Internet [1].

Industry 4.0 is currently among the leading terms in the world of business and organizations. The concept was presented at the Hanover Fair in 2011 and quickly attracted great attention not only in Germany but also worldwide. The Industry 4.0 concept is associated with the fourth industrial revolution in manufacturing, where technological trends such as digitalization, robots and artificial intelligence transform production processes [27]. A more accurate definition of Industry 4.0 is that it is "a new approach to manufacturing process management, providing real-time flow synchronization and allowing for unitary and individual production of products." The growth of Industry 4.0 is facilitated by government subsidies and attention, and Industry 4.0 quickly became a large and profitable industry in its own right. Technology providers (such as SAP), consulting firms (such as McKinsey, Boston Consulting Group), and conference organizers compete for market share associated with Industry 4.0 by offering and providing support and assistance services. The latest methods, such as artificial intelligence and robotization, the Internet of Things (IoT) and 3D printing, and virtual and augmented reality, are quickly becoming an integral part of our daily existence [16]. One of the main technologies of Industry 4.0, which is used to predict production processes, is the Internet of Things (IoT). With IoT, different personal devices and sensors can be connected to the network and transmit data on the state of production equipment, production lines and other production parameters.

In 2023, the global IoT market in industry totaled more than \$118.37 billion and will reach \$1.11 trillion by 2028. In Russia, by 2030, the volume of this market is expected to grow to 147.25 billion rubles. The Internet of Things is a complex of connected and connected to the internet and a variety of devices that exchange information. Similarly, the Industrial Internet of Things is designed to combine equipment, employees and cloud data [28].

With thousands of sensors installed in the enterprise and software, it is possible to automatically monitor production and control it: monitor the quality of products by choosing optimal modes or monitor the condition of equipment to predict its wear and tear and replace parts in a timely manner, extending its life and preventing downtime due to accidents. On the basis of data analysis, the load can be increased pointwise, thus increasing performance. Previously, statistical data collected manually or not collected at all were used for analysis. This makes it more difficult to access accurate information to assess the current state and forecast the future state of production. All of this can now be seen in real time via the Internet of Things. Today, promising development vectors such as smart cities and home systems, predictive diagnostic systems in industrial segments, robotics, medicine, and scientific developments in the artificial intelligence industry stand out in the IoT market in Russia [29].

The IoT is also actively used in the IT segment responsible for the software part of the hardware objects that are connected to the internet. Some IoT technologies simulate reality, test various digital scenarios and use a digital twin. The digital twin of an industrial enterprise is a copy of it in a virtual environment. The function of the digital twin is to reproduce processes and operations in the enterprise with high accuracy, which allows one to solve a wide range of business problems. In the philistine worldview, sometimes there is an erroneous idea of a digital twin as a 3D model, which is used to visualize the production infrastructure without any additional meaning or functionality. However, that is not right. The digital twin is, first, immersion in the logic of the processes taking place in the company, their digitization and, most importantly, the analytics of the data obtained via artificial intelligence. This is a promising technology for the future, and not a single large industrial enterprise in Russia in 10 years can be imagined without it [30]. The digital twin in an industrial enterprise provides accurate forecasting of production volumes and helps to determine the optimal amount of equipment for organizing the production process and the necessary reserves of production resources. The technology evaluates the peak and optimal performance of each unit in the production chain to predict the volume of contracting and helps to control the wear of equipment under various loads. Digital twins are also effective in the operation of industrial construction facilities. In addition, such solutions are used to assess the effects of the introduction of digital technologies and the commissioning of technological, engineering and

bench equipment. Depending on the specifics of the work of a particular enterprise and its development strategy, the functionality and interface of the twin can be changed to solve specific problems. A distinctive feature of the digital twin is that this asset can work for the company forever [28]. Against the background of the spread of smart devices and the development of data transmission networks, the interest of major players in the software market in SCADA systems (supervisory control and data acquisition), dispatch control and data collection) is growing. With the help of such solutions, it is possible to aggregate information from sensors and sensors for monitoring and controlling equipment into a single data lake and generate reports with visualized data for decision makers.

The manufacture of more complex devices requires the modernization of the production facilities of companies. The more complex the production is, the more complex the sensors and the more valuable the information they provide. Against the background of the active growth of data flows, the negative role of the human factor increases, and the risk of making incorrect decisions increases because a person cannot process all the information collected. The specified requirements are applicable both to commercial firms and nonprofit organizations. All organization should respond quickly and adequately to changes in the situation of society and the market and make appropriate changes in their activities [31].

The concept of Industry 4.0 represents a paradigm shift in industrial production, encompassing the agile generation and analysis of vast amounts of data in real-time. This development holds immense potential for enhancing strategic and operational decision-making processes. At its core, Industry 4.0 leverages data and connectivity to foster intelligent networks characterized by a high level of collaboration, ultimately contributing to a substantial boost in overall industrial productivity.

As illustrations of the implementation of lean manufacturing techniques and the integration of digital technologies, this paper provides examples that, when combined with advanced technologies such as big data, blockchain, and the Internet of Things (IoT), can significantly reduce downtime, enhance throughput, and improve overall efficiency. For instance, FANUC and Cisco have collaborated to develop streamlined systems that provide added value to manufacturing processes. The goal of achieving autonomous machinery in Industry 4.0 can be achieved through the use of blockchain technology, which connects an enterprise resource planning (ERP) system, a spare parts supply chain, and a cyber-physical production system within the factory. This enables machines to autonomously and reliably order spare parts as needed. Moreover, blockchain's capacity to facilitate transparent financial transactions among intelligent devices is crucial for the economic transformations brought about by the advent of Industry 4.0.

In order to bridge the research divide, the author proposes a hybrid approach that combines the principles of lean manufacturing with the tools of Industry 4.0.

The article, based on the initial improvement proposal, introduces concepts such as «lean manufacturing» and «Industry 4.0», which are widely employed in industrial settings. In this context, certain concepts have been further elaborated with specific use cases in mind. We have conducted a comparative analysis, examining the aspects of lean management that contribute significantly to the successful integration of digital technologies within the proposed model framework.

A conceptual framework for Industry 4.0, designed as a complement to Lean Production, should, on the one hand, provide recommendations on how Industry 4.0 solutions can reasonably support Lean Production practices. On the other hand, it should include illustrative examples. A comparative analysis of Industry 4.0 enabling technologies and Lean Production methods offers a comprehensive view of the potential synergies between them (Table 1).

2.3 Artificial intelligence development in “Industry 4.0”

To understand the evolution and application of artificial intelligence (AI) in an industrial setting, it is essential to explore its historical background and benefits. In this section, we will begin by examining the history of AI development and then discuss its implementation in this context. This section will also focus on the impact of AI and potential challenges that may arise during its implementation. Artificial intelligence is a significant technological advancement that contributes to the Fourth Industrial Revolution due to its impact on digital systems and business strategies. The concept of "artificial intelligence" was introduced by John McCarthy at the first conference on the topic, held at Dartmouth College in 1956. His vision was for machines that could think logically and acquire true knowledge by following instructions and providing appropriate responses. However, this approach presented challenges for programmers, requiring precise guidelines and definitions. As a result, artificial intelligence evolved from this traditional approach, with machine learning becoming a more popular strategy. Machine learning allows machines to extract insights from large datasets, gradually improving their capabilities. This method eliminates the need for carefully crafted programming instructions. This shift has led to groundbreaking developments in various fields, particularly in image recognition and natural language processing. Artificial intelligence refers to machines

Table 1 The potential applications of integrating Industry 4.0 and lean manufacturing practices

Industry 4.0	Lean production		
	Principle: just-in-time		Principle: Jidoka
	Method: Kanban system		Method: Andon
	Employee gets information about remaining cycle time via augmented reality		Wearable computing systems receive failures and display it in real time to the employee
Smart operator			-
Smart product	Smart Product contains information of Kanban to realize an order-oriented production		Machines send failures directly to Smart Operators and call other systems for fault-repair actions
Smart machine	Machines offer a standardized interface for receiving and sending Kanban		-
Smart planner	IT systems reconfigure production lines and update Kanban according to the new configuration		-

that exhibit intelligence through the use of computational resources and algorithms. These machines go through a learning process, gaining knowledge and improving their performance, leading to significant advancements in different areas. The emergence of AI has led to a significant change in the way industries operate, replacing manual labor and traditional methods with digital solutions and automation. AI allows machines to make intelligent decisions, leading to increased efficiency and innovation. The origins of AI can be traced back to the second half of the twentieth century, when automation and robotics began to emerge. However, it was not until the 2010s that a true revolution occurred with the advent of machine learning and deep learning technologies. These technologies have enabled industries to tackle complex problems and achieve remarkable success [32].

In terms of Industry 4.0 technologies, the development of artificial intelligence deserves a special focus. This is because automated systems require a unique approach due to several reasons [27, 28, 32]:

Firstly, within automated systems, one of the primary tasks is processing unstructured text data that is not in a natural language. This requires human-like intelligence to understand and interpret the data.

Secondly, natural language processing, among other AI technologies, faces the most challenges (especially in terms of identifying the meaning of text), and is at a relatively low level of technological readiness. This is due to the complexity of the tasks involved in semantic processing of natural language. As a result, an increasing number of researchers have concluded that it is beneficial to combine neural network AI with logical AI in order to create a universal (strong) artificial intelligence that can reason, autonomously extract knowledge from books and other documents, and solve previously unknown problems.

The latter circumstance, in our opinion, creates the potential for finding effective solutions, while also understanding the risks that may arise if the outcomes of such a combination of logic and neural network AI were to be obtained outside of Russia, for example in other countries involved in the development of strong AI (Artificial General Intelligence—AGI) [33].

In recent years, research has provided more and more evidence to support the idea of creating a universal (powerful) AI in the coming decade. This optimism is primarily due to the results shown by so-called baseline pre-trained neural network models (also known as foundation models). These models, primarily of the transformer type, undergo initial training (without supervision) on large amounts of text data measured in terabytes and containing hundreds of billions to trillions of parameters. As a result, they are able to rapidly complete their learning (with supervision) in highly specialized areas with relatively small training sets.

Artificial intelligence (AI) has the potential to automate the execution of intellectual tasks in various sectors of the economy and social sphere. However, there is an urgent need to objectively confirm the functional correctness of AI systems under specified operating conditions. This is a crucial prerequisite for the creation of a national system for assessing the conformity of products and services using AI technologies.

Currently, AI is successfully solving problems of intelligent data processing in areas such as transport, healthcare, industry, education, and agriculture. However, its high potential is limited by risks that can have economic, human, and state security consequences. Artificial intelligence technologies, including those based on machine learning methods, have a number of characteristics that are essential for their practical use.

These include [27, 28, 32, 34]:

- Incomplete interpretability, which means that it can be difficult to understand how the AI system works and why it makes certain decisions.
- Dependence on the quality of the training data (RD), which means that the performance of the algorithm depends on the accuracy and completeness of the data used to train it.
- The possibility of additional training during operation, which allows the algorithm to adapt to new situations and improve its performance.
- Transfer of responsibility for incorrect actions from the human operator to the AI system, which can help reduce human error.
- Need for comparing the performance of AI algorithms with human capabilities, to ensure that they are effective and reliable.

However, the practical implementation of AI is limited by several factors, including the lack of guarantees for physical and information security, as well as social acceptability and consumer satisfaction. The provision of the aforementioned guarantees is ensured by confirming the functional correctness of SI: the degree of ensuring the necessary accuracy of correct results under specified modes and conditions of use. At the same time, the consequences of practical use of AI

with unconfirmed functional correctness can be divided into three categories, depending on the severity of the consequences from the occurrence of a possible event.

Minimizing the risks associated with the practical application of AI technologies would allow the creation of a national compliance assessment system. Voluntary certification, as a form of compliance confirmation, is carried out independently by various stakeholders and includes product testing, ensuring the greatest objectivity in the results obtained. To date, there are two main approaches to creating requirements for products that use AI technologies and assessing their compliance with these requirements. These approaches have been developed in relation to vehicles. One approach considers AI as a structural feature of the product, which means that the development of specific requirements for AI systems is not necessary [33].

As we have previously discussed, this technology is intersectoral in nature and allows for the unification of requirements for verifying the functionality of AI algorithms and coordinating the development of evaluation methods and tools. This helps to avoid duplication of effort and ensures that uniform standards are followed when creating test data sets.

In light of this, we can now discuss the development of AI and its benefits. Based on this, let us consider a definition that is relevant to the context of our discussion. A technology that enables automata and robots to make decisions is a significant advancement. It gives machines the ability to collect information from their environment, process it for analysis and decision-making, learn from experience, and solve increasingly complex problems. This technology has the potential to transform various aspects of human life.

Considering the term we have presented, it is important to note that challenges arise during the implementation of this technology in enterprises. To address these challenges, we turn to the practical research conducted by Sinha and Lee [34]. They propose to identify these challenges and offer strategic recommendations for the application of this technology in various industries.

Considering the presented term and the research example, it is important to take into account the factors hindering the development of this technology. In this regard, we should consider the list of problems proposed by the authors, who believe that these are the main challenges of artificial intelligence in various industries. It is worth emphasizing that these problems can be solved by implementing lean and digital management methods using artificial intelligence in an industrial environment.

Deploying AI solutions in industry is a complex process that involves technical and operational challenges. It requires a deep understanding of both the theory behind AI and the practical realities of industrial environments. In addition to technical difficulties, there are a variety of logistical and human factors to consider during the deployment process. These include managing issues such as data privacy, security, and regulatory compliance. Figure 1 provides a list of some of the challenges associated with deploying AI applications in industrial settings.

Based on the list of issues presented, we can draw some key conclusions about the development of artificial intelligence (AI) in industry. The process requires attention to several aspects, from managing data to choosing an appropriate algorithm for the system. The deployment of AI systems brings its own set of challenges, including ensuring system compatibility, scalability, and addressing critical security concerns.

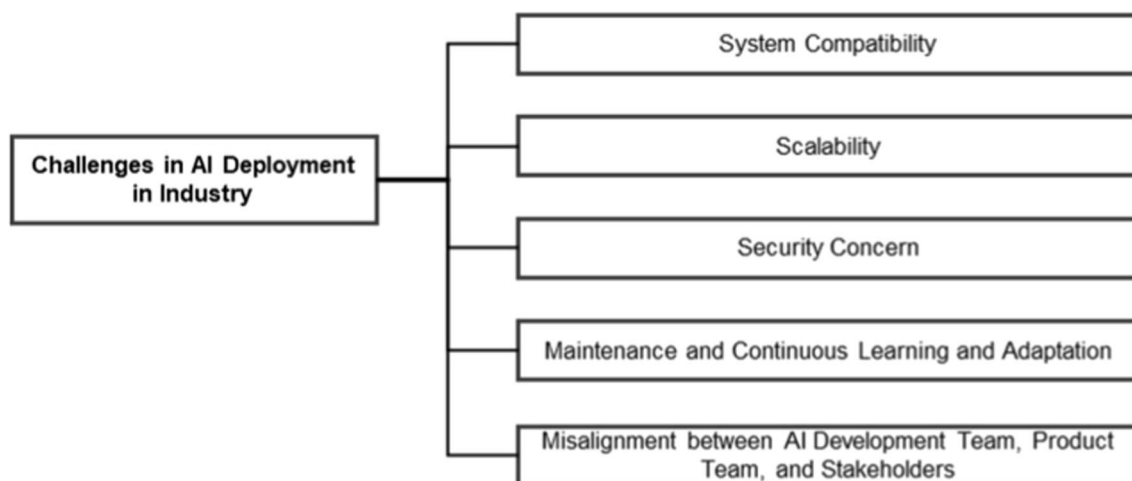


Fig. 1 List of challenges in deployment of AI applications in industry (Sudhi Sinha, Young M. Lee)

In addition to these technical considerations, integrating AI into industry also raises ethical and regulatory concerns. Concepts like bias, accountability, transparency, and data privacy are important issues to consider in this context. These topics may become a focus of future research and discussion.

3 Research design

3.1 Optimizing the implementation of digital technologies in the context of Industry 4.0

Therefore, to leverage the potential of digital technologies (DT) in existing lean-based production systems (LPSs) [29–31, 35], manufacturing companies need to redefine and reconfigure their organizational strategy, structure, infrastructure, resources, and culture [36, 37].

More specifically, two research questions are posed:

RQ1: What concepts and technologies allow businesses to take an integrated approach to problem-solving in the digital world?

RQ2: What strategies can managers develop that can be seamlessly integrated into the work of an organization to use digital production technologies with artificial intelligence to achieve its goals?

Considering the questions raised, we will move forward with this solution through an optimization process aimed at enhancing the operation of the company and its processes. Implementing Industry 4.0 technologies optimally involves developing and implementing methods and strategies to maximize opportunities provided by new digital and automated lean management technologies.

The goal of this process is to enhance the efficiency, quality, and competitiveness of the company by utilizing advanced technologies and improving production processes. This includes analyzing and optimizing current production procedures, deploying artificial intelligence algorithms, implementing production management systems, and managing customer relationships effectively. By doing so, businesses can improve their overall performance and remain competitive in a rapidly evolving business environment.

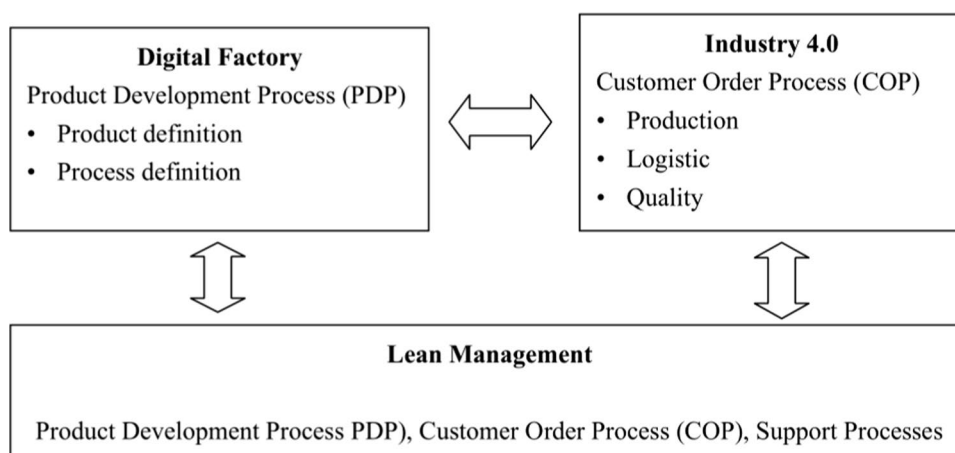
Thus, intelligent enterprises and advanced manufacturing facilities integrate a variety of components (Fig. 2), some of which are developed in-house, while others are acquired through competitive acquisition. These elements can be interconnected, and the concept of Industry 4.0 cannot be fully realized without establishing the fundamental principles of lean manufacturing, including process identification and measurement [35].

Sanders, Subramanian, Redlich, and Wolfsburg proposed that the synergistic effect arising from the interaction between Industry 4.0 and lean manufacturing can help us achieve our goals [36].

The present research has unearthed five publications that corroborate this matter. Mayr et al., in their work published in 2018, delved into the question of whether lean management and Industry 4.0 are mutually supportive [37].

Thus, as other studies have shown, the link between Industry 4.0 and lean manufacturing is extremely important to ensure the quality and reliability of products. In addition, there is a high level of synergy between these two concepts.

Fig. 2 Optimization approaches and their underlying technical processes [35]



Dombrowski and Richter confirmed this point of view by reporting a high correlation between cloud computing, waste prevention, defect-free and big data [38].

When comparing the features of Lean Management (LM) and Industry 4.0 (Industry 4.0), it becomes evident that there are certain parallels between the two approaches [36]. The five core principles of LM as outlined by Womack et al. in 1990, along with the "people" principle proposed by Hines et al., appear to align closely with the fundamental competencies of Industry 4.0 as described by the Bitkom [39].

3.2 Integration of digital technologies via lean manufacturing

One of the primary objectives of contemporary domestic research in the field of digital transformation is the development of scientifically grounded solutions for the streamlined operation and enhancement of production processes within enterprises. This encompasses the widespread implementation of digital technologies and the establishment of practical planning methodologies across all levels of industrial organization.

This reasoning highlights the fact that the combination of Lean Manufacturing and Industry 4.0 is still being studied and referred to as Lean 4.0. Rittberger and Schneider used the Human-Technology-Organization model to compare and integrate Lean manufacturing with Industry 4.0 in order to identify potential opportunities. Their research shows that the Lean manufacturing approach focuses on organizations, people (such as employees and customers), and their interactions. They show that the lean production approach focuses on organizations and humans, such as employees or customers, and their interactions [37, 40, 41].

The research conducted by the authors highlights the key advantages of Industry 4.0. Primarily, the digital platform fosters the generation of new employment opportunities and attracts investments. This leads to an increase in productivity, enhancement of product quality, reduction of risks, and optimization of resource allocation [28–30].

It is worth noting that the term "Industry 4.0" often refers to the fourth industrial revolution, which involves the integration of cyber-physical systems into production processes. Let's take a look at the situation in Russia. In general, digitalization in Russia's industrial sector has gained momentum, with our share of the global economy at just over 22%. This is not a bad position, and Russia leads in terms of innovation: 7% of companies are leading, 19% are innovative, and only 1% are lagging behind. However, there are 9% of European and global companies that are leading in digitalization. Approximately 40% of Russian industrial enterprises plan to transition to digitalization over the next five years, indicating a positive trend [42].

The implementation concepts need to be well aligned with the methodological approach to managing the production system when the concept of lean practices is implemented.

The combination of lean production and Industry 4.0 is still being researched and is called lean 4.0 [35, 36, 38].

Moreover, it is a holistic and long-term-oriented methodical approach that requires a change in employees' attitudes and awareness to reach a continuous value stream by avoiding waste [43, 44].

Moreover, Industry 4.0 is technology focused and requires a high degree of process orientation with clearly defined processes [40, 45].

It covers the interface of technology with both humans and the organization [46].

Considering the analysis and the critical approach, it is important to note that the introduction of lean manufacturing requires a well-coordinated methodological approach to manage the production system. The combination of lean manufacturing and Industry 4.0, or "lean 4.0" as it is also known, is still in the research and development phase. This holistic and long-term approach necessitates a change in employee attitudes and awareness in order to ensure a continuous flow of value creation and loss prevention. Industry 4.0 emphasizes technological and process orientation, necessitating well-defined procedures. It also entails the interaction between technology, people, and the organization.

Any manufacturer wants to achieve maximum results from their production. To this end, companies from various industries invest in methods for managing and optimizing their technological processes. Within the context of digital transformation, the need for reliable equipment unites enterprises. Perhaps this is a crucial aspect of digitalization. Regardless of what product the company produces, equipment malfunctions always lead to losses and downtime [47].

In addition, if a company uses digital technologies to predict and prevent equipment malfunctions, this would be a significant advantage for the company.

Concerning the first perspective, many scientific papers state that lean production is the basis for Industry 4.0 [48, 49], as Industry 4.0 technologies are assumed to be built on the holistic approach of the lean production system [50].

4. In the course of modelling the examples, I meticulously analysed the feedback and recommendations provided. Building on the fusion of lean manufacturing principles with the advent of Industry 4.0 technologies, I propose adopting

a model that acts as a foundation for the seamless integration of lean production practices and digital advancements into industrial workflows. This model is a meticulously structured and comprehensible example.

Given the above-mentioned examples of implementing the Kanban method in lean manufacturing in the context of Industry 4.0, it becomes necessary to take into account the potential of interfaces that can facilitate the smooth integration of these digital technologies into existing systems. Similar to service-oriented architecture (SOA) in lean manufacturing, cyber-physical systems (CPS) are capable of providing basic services to both on-premises and higher-level systems, as shown in Fig. 3.

Thanks to this, workstations can seamlessly integrate into production lines, executing commands from an advanced manufacturing management system. Moreover, CPS is capable of exchanging data with sensors through the Internet of Things (IoT), actuators, or programmable logic controllers. Additionally, it enables interaction with operators at the workstations through a human-machine interface. This allows for the upgrading of previously manual workstations with automated technologies, as well as vice versa, without the need to modify the manufacturing management system. CPS guarantees the autonomy of workstations from IT systems and their operators.

The basis of this model includes the fundamental components. This structure is a description of the functions of the devices and their responsibilities, as well as the sequence of certain tasks. In the context of lean manufacturing, every workplace should be able to receive kanban, interpret it, and initiate the production process. These tasks do not depend on the configuration of a particular workstation, whether it is an automated system or manual control. In addition, any malfunctions at the workstation must be immediately reported to the responsible employees, as well as an improved continuous improvement system.

A survey by Tortorella & Fettermann confirmed that Industry 4.0 is more likely to be introduced if lean production is already implemented [41].

An essential requirement of Industry 4.0 is a certain degree of process orientation with defined processes, customers, suppliers, tasks, and times [38]. In support of this, lean production system generate standardized, transparent, and waste-free processes [51, 52].

The challenges faced by project managers in enterprise development and implementation are more complex and dynamic than ever before. To address these challenges, we conducted a study focused on two key areas identified during

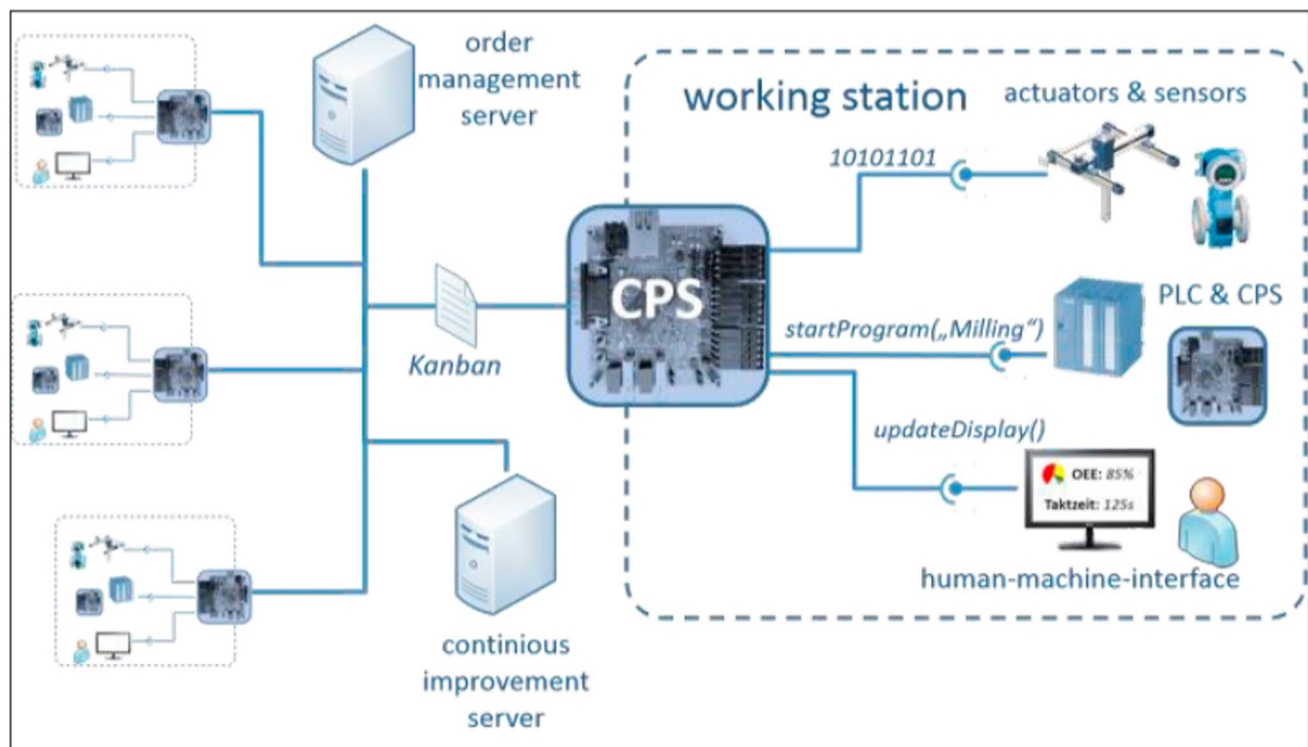


Fig. 3 A framework for the harmonious integration of lean manufacturing principles and digital technologies within the realm of industrial production

the design process. This document presents a comprehensive methodology to address these issues. It is divided into two main sections corresponding to the two stages of the research:

In the first part, we will describe the pilot project and its objectives, as well as explain how the introduction of modern technologies, including artificial intelligence (AI), allows businesses to take a holistic approach to problem-solving in the digital world. We will also discuss how these concepts and technologies relate to the first research question (RQ1).

In the second part, we will present the results of the project and discuss how modern concepts such as lean digital production have enabled us to achieve our goals (RQ2).

The outcome of our research is a model that managers can use to seamlessly integrate digital manufacturing technologies and AI into their organization's workflows to achieve their objectives. The result of our research will be a proposed scheme based on visual principles and the use of digital tools in real time to make informed decisions. Our work aims to implement digital transformation projects that will allow the company to take advantage of digital tools and solve problems. We are also changing our approach to production processes and behavior, considering the recommendations provided by data-driven analytical tools. These technologies will allow us to design, create, and test new business models. Based on the research positions mentioned above and the discussion points discussed regarding our research, we would like to move on to the presentation of our results.

4 Results

4.1 Implementation of lean and digital manufacturing projects: use of artificial intelligence

As a first result, we will describe the process of the company's work when switching to digital production. We will look at a well-known process within the enterprise. Let us consider a well-known process at the enterprise. At the beginning of the process, data are collected from sensors. Then, their complex processing by algorithms via software is carried out, taking into account processes and production. Furthermore, to ensure the correct operation of the entire circuit, the actuators are immediately activated. Enterprises have other technological processes that do not fit into the framework of such a closed cycle. The amount of process data can be limited, information collection can be carried out manually, and the analysis is practically not carried out. An example of such a process is monitoring the reliability of the equipment. Monitoring the security of processes, employees, and energy management. Using artificial intelligence in this case, we understand that it is necessary to close the circle or automate all these areas (Fig. 4).

This diagram shows that in order to make informed decisions, we need accurate and reliable data in real-time. This data can come from new measurement methods or entirely new data collection techniques, such as sensors using artificial intelligence. The data is then transferred to a computer or application, or to an employee, anywhere in the world, via the internet. It is then analyzed using digital technologies, such as digital twins or analytics with machine learning, or with the help of employees or experts. Thanks to cloud storage, there is almost unlimited potential for analyzing, storing, and processing this data using analytical tools. All this work aims to implement digital transformation projects. Thanks to digital tools, people can work remotely, even if the data they are working with is located far away. Secondly, we are changing the way we approach production processes and behaviors, taking into account data and recommendations from analytical tools. These technologies allow us to design, create, and evaluate new business models. Traditionally, employees in manufacturing companies have been involved in optimizing technological processes. They gather data independently and use software to process it or involve their own engineers to solve problems. However, in the past ten years, a number of remote locations have emerged, including test centers, landfills, mines, and oil and gas production platforms. This has led to the idea of a centralized control center for integrated or "digital" production. It has become clear that experts can be gathered in one central location where all data can be collected and shared, instead of having them travel to different sites. However, this still happens within the information systems of a single company and is supported only by its employees. Due to the digital transformation and the industrial internet of things concept, the company has adopted a new approach to services. The presented system utilizes artificial intelligence (AI), which allows innovative products to learn, adapt, and continuously improve. Digital devices gather information from their environment and interact with each other to accelerate their development. The ability to transfer knowledge is impressive and remarkable, as it allows experts to share their knowledge on various topics and issues.

Within the framework of this work, the scheme of the cycle of optimization of the production system using artificial intelligence in enterprise management was considered. In the course of the study, well-known methods were used to determine the standard time for performing equipment setup operations by the authors Tin-Chi Toli Chen, Chi Wang [53].

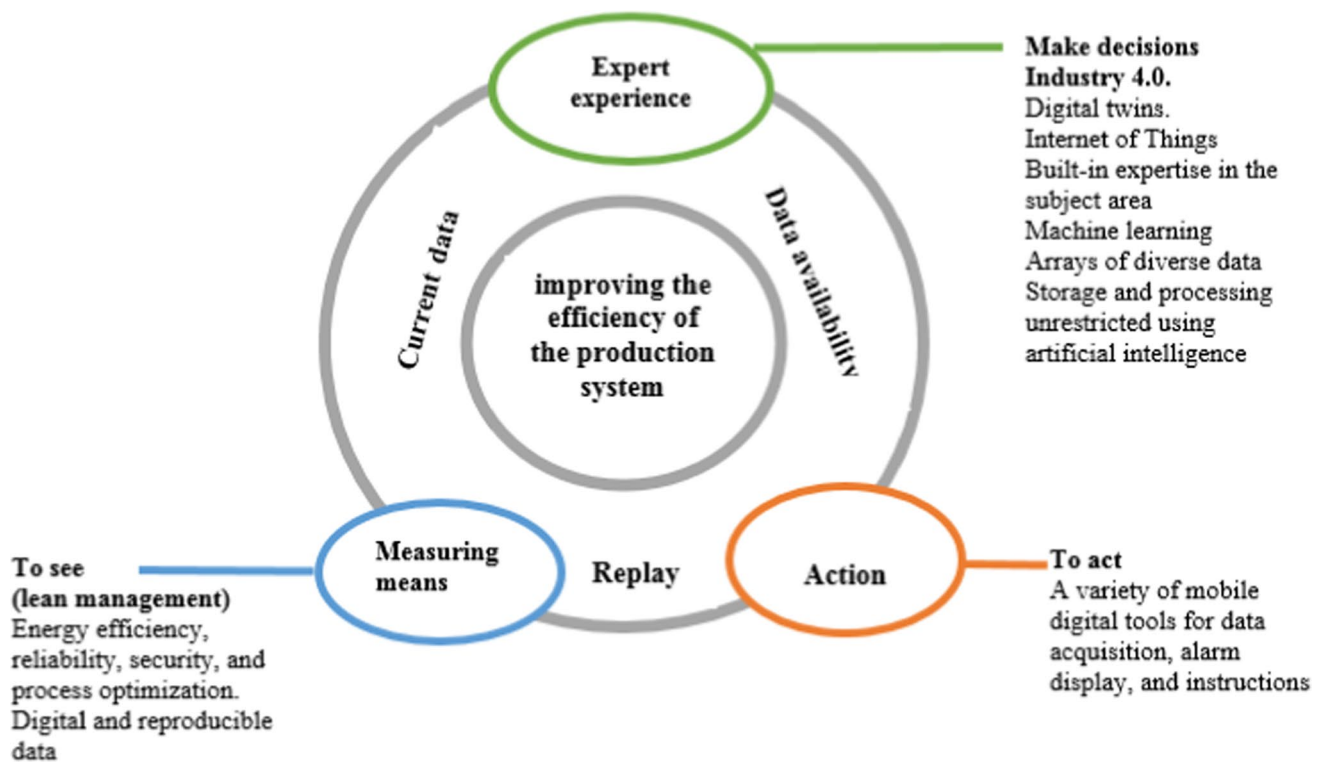


Fig. 4 The scheme of the cycle for improving the efficiency of the production system using artificial intelligence in enterprise management

In order to establish the standard duration of a novel setup procedure, a comprehensive analysis of ten analogous operations was conducted, as outlined in Table 2. A neural network (ANN) with the architecture depicted in Fig. 5 was developed to forecast the standard time required for a setup operation denoted as a_j , based on its four characteristics $\{x_{jk}\}$. With the exception of the number of setups performed per day, all other attributes were derived from expert assessments, which yielded integer values. As part of the study, data was collected using sensors, which were subsequently processed using artificial intelligence algorithms. All of the collected data are used to train the ANN, for which the training algorithm is the Levenberg–Marquardt (LM) algorithm [53–55].

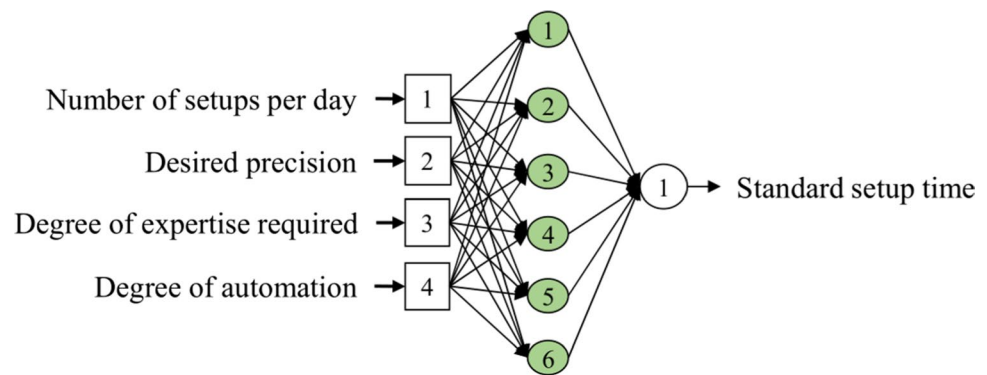
The results of the forecasting process have been summarized. The accuracy of the forecast, measured in terms of root mean squared error (RMSE), is 0.134 min. The values for the four parameters of the new configuration are 9, 2, 4 and 3 respectively. Following the application of the trained artificial neural network (ANN), the standard setup time has been determined to be 9.6 min.

Transfer learning is a method that utilizes a pre-existing model to solve new problems. This approach saves time and resources by avoiding creating a model from scratch, and it also allows for scaling up AI by applying knowledge gained from

Table 2 Data of some setup operations

j	x_{j1}	x_{j2}	x_{j3}	x_{j4}	a_j (min)
1	10	3	3	2	19
2	10	3	1	3	16
3	9	3	4	4	20
4	5	4	3	1	25
5	19	1	1	3	12
6	3	5	2	4	29
7	4	4	2	4	22
8	10	3	3	4	19
9	8	3	1	4	15
10	8	3	4	4	20

Fig. 5 ANN for establishing the standard time of a setup activity



one problem to similar ones. Connected learning further improves this process by retaining knowledge within the model, reducing costs, and increasing efficiency [32].

Based on the outlined framework, several critical insights can be derived. The proposed framework is groundbreaking and offers several advantages over current approaches.

In the digital realm, artificial intelligence (AI) is now integrated into products, enabling them to continuously learn, adapt, and evolve. Digital devices acquire knowledge from their environment, accelerating their development through data exchange. The capacity to share knowledge is both remarkable and potent.

Our framework employs transfer learning, a neural network technique that transfers knowledge from a pre-trained model for a specific task to another related task within the same domain. Connected learning retains knowledge within the machine learning model, substantially reducing the cost of developing new models and enabling economies of scale in AI applications.

From an economic perspective, Huber expects a more efficient implementation of Industry 4.0 technologies to occur when lean manufacturing is already in place [36]. Additionally, technological advances in Industry 4.0, such as artificial intelligence, have the potential to complement or even enhance lean manufacturing processes, leading to increased effectiveness. The other central perspective is that technological advancements in Industry 4.0 might complete or even enhance lean production activities and positively affect their efficiency [37, 56, 57].

Bauernhansl and Kieviet highlighted the potential of Industry 4.0 for lean production systems facing complexity and flexibility simultaneously by applying appropriate technologies [58–61].

In this case, the company can attract remotely and use digital outsourcing, i.e., digital services of experts of a third-party organization, which means that it is not necessary to involve its own employees for analysis and data processing. This is a very effective business model that solves the problem of organizing and performing work.

To use the domestic potential of digitalization in full force, it is necessary to follow these recommendations [28–30].

1. Avoid excessive haste, which carries significant risks. You cannot instantly translate all production into digital.
2. Forcing digitalization with a low level of information security is dangerous.
3. Gradual reforms are needed, and you need to start with less.
4. The digitalization of production and the economy involves nothing more than very large costs. Therefore, it is necessary to stimulate investment, create a regulatory framework and, most importantly, take care of digital literacy and the development of the digital competencies of workers.

Let us draw the line of the above: the introduction of digital technologies has certain advantages, but excessive haste and forcing digitalization carry serious risks. It is preferable to gradually carry out activities.

Therefore, this scientific contribution provides a systematic approach to identify the implications of lean production and Industry 4.0 methods on target dimensions that are highly important for the development of production system enterprises.

4.2 Implementation of the lean manufacturing method using the 5S visualization system and the integration of digital technologies

As a second result, we would like to describe a pilot project that integrates digital lean manufacturing, using the example of a production site in our workshop. The aim of the project is to introduce lean manufacturing methods through the use

of the 5S visualization system and the implementation of digital technologies. By implementing this project, we aim to achieve our ultimate goal - to enhance the performance of our enterprise.

We will describe the main methodological approaches to implementing the system. Workplace order must comply with 5S system standards. It is essential to ensure compliance with workshop 5S standards. At the first stage, sorting, employees must determine what is needed for the production process and what needs to be discarded or used elsewhere on the premises. According to workshop employees, sorting is time-consuming and challenging. This step is necessary as, according to instructions for implementing the 5S system, employees should identify all unused materials, equipment, and other items accumulated in the workshop over years. Some of these items may be significantly obsolete and not currently used in workflows [19, 26].

Several issues were identified during the project's implementation. Some products were not labeled or identified, and none of the workshop workers knew exactly what they were or how they would be used in the production process. Based on available data, 24 products from the first stage of production remained unused after the initial stage's completion. Due to these unnecessary items, valuable resources were wasted, which could have been used to store essential tools and equipment for installing truly necessary equipment. In addition, all these extra and unnecessary items distracted workers and wasted valuable time. To solve this problem, we took measures that led to positive changes in the workshop's operation. Currently, all necessary tools are neatly organized in drawers and shelves, making them easier to access. To automate the tool storage process and reduce search time, we introduced automation of the accounting tool storage system in the warehouse. This has significantly reduced the time and effort spent on searching for tools and organizing production. For a more comprehensive solution, we analyzed the production process and identified six stages. Each stage was analyzed individually to identify areas for improvement.

The production plant's range included the production of packaging for the chemical industry, which was used for large-scale transportation together with the goods being transported. The subject of analysis was a production process that consisted of six stages performed at separate, individual production stages (Table 3).

The production process of containers for transporting products begins with the collection of the necessary elements from the warehouse of raw materials. Next, operations 10–40 are carried out in turn, as outlined in Table 1. Simultaneously, the second process of preparing a batch of operations 50 takes place. The finished packaging, created in accordance with the production order and matching the dimensions of the transported goods, goes to the interpretation buffer. Finally, in operation 60, the product is properly packaged in the manufactured container. The standardization process involves creating instructions for individual production operations and organizing the sequence of actions during the preparation phase of the manufacturing process. Since the implementation of 5S and standardization tools, regular checks have been carried out to ensure the correctness of the procedures implemented. This confirms the successful implementation of these tools (Table 4).

Time measurements of operations were carried out every day using the working-day photography method. The personnel performing production operations often rotate between different production positions. Another important principle of the 5S system is the use of visual aids and associated notation. This also allows you to save time, quickly obtain the necessary information, and act strictly in accordance with the instructions.

For example, workshops in recent years have been equipped with a large number of new machines. Each machine contains information about who is responsible for it and who works on it. Additionally, the cost of the machine is indicated. Why is this information provided? What does it have to do with lean manufacturing?

It turns out that it has a lot to do with it. We have noted that lean manufacturing is designed to conserve resources. Moreover, a machine, which is all the more valuable, is a resource. In this case, we are talking about increasing employees'

Table 3 Description of the operation

No. operation	Description of the operation
10	Retrieving components and assembling the base of the structure x
20	Preparing the components and making the frame of the structure
30	Making screw connections of components
40	Preliminary preparation for the operation
50	Preparation of the transported product for packaging
60	Packing the finished product in accordance with the production order and securing the product

Table 4 Improvements introduced as part of 5S and Standardization

No. operation	Description of the operation
10	Limiting the number of structural elements used during operations at the production station to only those used during the production order
20	Preparation of construction instructions for a specific (implemented at the moment) production order
30	The use of a shadow table and the limitation of tools at the production station
40	Changing the structure of the station allowing for easier execution of operations and marking the place of transfer of the finished product
50	Preparation of work instructions defining unified standards for product preparation for transport to the interpretational buffer
60	Implementation of structural changes to the station facilitating the implementation of operations. Standardized work activities-workplace instructions

responsibility to prevent carelessness or negligence in machine failure. This measure is also designed to directly involve personnel in the process of reducing costs, losses, and increasing production efficiency.

The introduction of new methods for organizing the production system requires a restructuring of the way we think and approach business, first by managers and then through the involvement of all workers. Sorting was carried out not only in common areas where equipment was stored, but also at each individual workstation. After the sorting was completed, a rack was installed in mechanical section No. 1 and systematization was done, designating large equipment in the workshop. The 5S system is implemented at workstations on an ongoing basis by site supervisors. Every day before the work shift starts, when the shift-day tasks are issued, operational control is conducted with site operators, tasks are assigned, and schedules are checked. The new approach to production assumes that the operator has a clear understanding of the goals and objectives of their site, and the manager is able to monitor task completion, identify deviations, and take corrective action in real-time [26].

As a result of the Lean project, we have been able to receive real-time data on the availability and location of each tool and piece of equipment. This has allowed us to create a forecast of tool demand with an accuracy of up to 5%, and maintain a stock of tools at work centers, warehouses, and storage facilities at a level of 10% of projected demand. Thanks to digital technologies, we have also reduced the time spent preparing and searching for tools during the month, and the losses associated with these activities [43].

5 Discussion

Summing up, it can be stated that this article makes a significant contribution both to the scientific community in terms of theory development and to the business sphere in terms of practical implementation. Previous studies have identified the advantages of integrating digital technologies into traditional logistics process management systems. More recent research has focused on the implementation of the principles of Industry 4.0, which include the use of well-known principles of lean manufacturing. These principles, such as the 5S system, Kanban, Kaizen, TPM, JIT, SMED, TPM, Poka-Yoke, U-shaped cells and visualization, are integral components of this approach [10, 11].

The research we reviewed in this article demonstrates the benefits of integrating digital technologies into existing enterprise management systems [12, 15–17]. We also explored the implementation of Industry 4.0 using advanced artificial intelligence techniques in an industrial setting. Our research has shown that it is essential to prioritize objectives in order to achieve success. To do so, it is important to establish regular communication channels throughout the organization. This includes those who are leading the transformation and those who use the system on a daily basis. This approach to fostering communication and collaboration differs from traditional practices in financial institutions. However, cooperation remains a crucial aspect of any digital transformation effort [41].

The author's concept, developed through integrated cooperation, allows for the systematization of theoretical and methodological principles in enterprise management, as well as the improvement of organizational changes aimed at making management decisions [14, 15], or [20].

The results of the study are consistent with data on other manufacturing processes described in the literature. By examining these results in more detail and comparing them with data from other manufacturers, we can better understand why lean manufacturing (LM) methods are becoming so popular [67, 68]. Additionally, we will be able to evaluate

their impact on production in the context of Industry 4.0. Given that methods such as 5S and standardization, which are already widely used in many companies, have been found to be effective, the findings from this study may encourage other companies to adopt solutions based on similar principles [19, 26].

The author has developed a theory that constitutes a comprehensive approach to the classification of theoretical and methodological foundations of corporate governance. Within this framework, he has addressed all the contentious aspects. This theoretical construct serves as a roadmap for organizational transformation, facilitating the enhancement of decision-making processes in the context of implementing lean manufacturing practices and integrating artificial intelligence into industrial operations.

In accordance with the previously discussed concepts, Industry 4.0 and the related technologies of the Fourth Industrial Revolution represent a broad spectrum of approaches that extend beyond the traditional frameworks of connectivity and digital automation in manufacturing. These technologies aim to enhance production efficiency and adaptability through self-monitoring capabilities and automated data exchange among components.

Moreover, enterprises can reap substantial benefits from the deployment of AI-powered processes. These processes appear to be more flexible and scalable than their conventional counterparts, enabling them to seamlessly integrate with a multitude of digital enterprises. This fosters a culture of continuous learning and improvement, transforming manufacturing processes and market interactions. The Fourth Industrial Revolution empowers companies to reinvent their operations and redefine their approach to market engagement, departing from the status quo [32].

This is achieved through a variety of techniques, encompassing big data analysis, the Internet of Things (IoT) integration, cybernetic systems, computer networking, robotics applications, collaborative robotics systems, and artificial intelligence integration. Industry 4.0 represents an evolving domain that necessitates a comprehensive understanding of its diverse aspects. Addressing the specific challenges within this field requires a deep comprehension of the underlying concepts and research domains. Such knowledge serves as the foundation for future research endeavors and development projects [62].

5.1 Extensive discussion of restrictions

Within the framework of this discussion, we shall explore the potential synergies that arise from integrating AI and lean manufacturing practices. Delving into the realm of artificial intelligence reveals a multifaceted landscape encompassing a multitude of technologies, offering vast opportunities ranging from decision-making algorithms capable of learning from extensive datasets to robots equipped with a degree of autonomy and manipulators for interacting with their surroundings. The extent of tasks that can be undertaken with AI assistance largely depends on the sophistication of the employed technologies, the scale of production capabilities, and the unique requirements of each enterprise. Small-scale businesses in particular are attracted to AI solutions that incorporate machine learning algorithms, driven by their rapid product line and technological process evolution, necessitating their capacity to address novel challenges effectively [53].

In the context of lean manufacturing, AI is viewed as an adjunct tool in the pursuit of lean manufacturing practices, possessing the ability to integrate with other tools and amplify their collaborative impact. The benefits of this integration have been explored through research, revealing its advantages for various industries. For instance, in the realm of global retail, the application of AI has been shown to reduce stock formation time by 30%. In the energy sector, AI contributes to a 20% increase in electricity production. In industrial production, it expedites material delivery by up to 30%. Moreover, in healthcare, AI enhances productivity by between 30 and 50% [54, 63].

One of the key management challenges where artificial intelligence (AI) can be effectively deployed is the orchestration of product launches, particularly in the realm of small-batch and diversified manufacturing. Beyond addressing the issue in its traditional form, AI also finds utility in the implementation of lean manufacturing practices, such as Just-in-Time approaches. In today's landscape, customers and consumers exhibit a preference for immediate product availability. Rather than waiting for an entire batch to complete processing, they seek to receive their orders promptly. Leveraging technological solutions that enable easy reconfiguration, AI can facilitate a hybrid approach to product launches. By incorporating AI-driven calendar planning, diverse products can be manufactured on a shared production line, ensuring adherence to the schedules devised by AI. The implementation of lean manufacturing tasks is based on a comprehensive analysis of the current situation. In such a context, the full potential of AI becomes indispensable. Moreover, AI plays a crucial role in the implementation of lean principles, enabling the resolution of a multitude of additional challenges. The resolution of these challenges contributes to enhanced labor productivity within the enterprise [55, 64].

Before delving into how AI assists manufacturers in adopting a lean approach to manufacturing, it is crucial to comprehend the intricacies of the limitations inherent in employing AI. During the monitoring process, raw data generated

by production lines is subjected to analysis. However, in conventional AI, unsupervised machine learning algorithms are used on raw data, resulting in a multitude of limitations and erroneous conclusions. In contrast, process monitoring AI contextualizes data by integrating business data from IT systems with a specific production context, thereby creating a model of the manufacturing process.

At the same time, AI uses machine learning algorithms capable of filtering out noise from data, leading to false positive results. This allows production teams to accurately identify targeted actions. Machine learning enables process and quality engineers to obtain predictive information, making it easier to solve problems instead of just conducting research. Users provide feedback to the machine learning algorithm in the form of accuracy and relevance ratings. This allows them to improve the accuracy of the algorithm over time. Human-machine cooperation aims to improve industrial artificial intelligence results. One of the primary applications of AI is to conduct analysis and make predictions about losses, leading to their reduction and improvements in quality [65, 66].

Through the use of AI, process engineers are able to anticipate and avert production waste by pinpointing areas of inefficiency and implementing targeted measures aimed at eliminating defects and enhancing product quality. This is accomplished through the application of predictive analytics and automated root cause analysis, which enable the prediction of process failures that give rise to wastage.

The use of artificial intelligence applications does not lead to job reductions in any way. Rather, their implementation can be structured in such a manner as to supplement the lean manufacturing process. This enhances the capacity of personnel by granting them access to real-time data and expanding the toolkit for analysing and executing production tasks.

6 Conclusion

In conclusion, we would like to highlight a few key points from our research. The study we conducted has identified gaps in the current knowledge of a particular field and allowed us to answer the questions we raised through scientific reasoning and practical experience. Our proposed methodological approach is based on scientific principles and comprises two main components: expanding theoretical knowledge in the field and developing methodological and practical techniques for implementing Industry 4.0.

The first component, expanding theoretical knowledge, aids us in gaining a deeper understanding of the subject matter and discovering new insights. Through this, we can identify the potential advantages of artificial intelligence and its implications for the industry. By comprehending these benefits, we can enhance our skills and improve our mutual understanding. The second aspect, the practical application, suggests new methods and strategies for incorporating Industry 4.0 into existing systems. This involves the use of artificial intelligence and transfer learning, which can aid in solving problems and achieving goals. By combining these two elements, we aim to demonstrate the practical implications of our research and advocate for implementing our approach in managing production systems. Additionally, the study considers a practical example of an organization's journey towards digital transformation through the implementation of Industry 4.0 technologies. In order to enhance operational efficiency and address the issues discussed in the research, the author proposes a model for enhancing the production system by integrating digital components, such as artificial intelligence. This integration has enabled the achievement of desired goals.

In general, the research results obtained represent an assessment of the feasibility and desirability of implementing a proposed cycle for improving the production process through the use of digital technologies within a company. This approach has clear benefits due to its evolutionary nature, as it does not replace existing control systems but rather complements them with practical methods for process control. Additionally, business leaders gain a better understanding of the need to integrate Industry 4.0 technologies, such as the Internet of Things and digital twins, which significantly facilitates their involvement in process management [69, 70]. This approach contributes to the development of procedures that cater to the diverse needs of industrial enterprises and reflect the efforts of many generations, representing significant intellectual value. An integrated management approach, incorporating lean and digital manufacturing as well as artificial intelligence and machine learning, provides a reliable path for enterprises to transition to digital methods. This approach can be applied in various business sectors.

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Declarations

Ethics approval and consent to participate The views and opinions expressed in this paper are those of the author and do not represent the official position of UL Solutions.

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